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Course: Game Programming

Assignment 01

Q1: What are the conditions for two colliders to get detected in OnCollisionEnter?

For OnCollisionEnter to fire, the following four conditions must be met:

1. **Colliders:** Both GameObjects must have a **Collider** component attached.
2. **Rigidbody:** At least one of the GameObjects must have a **Rigidbody** component.
3. **Is Trigger:** Both Colliders must have the "**Is Trigger**" checkbox **unchecked** (False).
4. **Kinematic Status:** At least one of the objects must be a **non-kinematic** Rigidbody (Dynamic) if you want the physics engine to calculate the impact automatically.

Q2: Difference between OnCollisionEnter and OnTriggerEnter?

- **Physical Interaction:** OnCollisionEnter handles physical impacts where objects bounce or block each other. OnTriggerEnter allows objects to pass through each other like a ghost (used for zones or pickups).
- **The "Is Trigger" Setting:** OnCollisionEnter requires **Is Trigger** to be **OFF**. OnTriggerEnter requires **Is Trigger** to be **ON** on at least one collider.
- **Data Provided:** OnCollisionEnter passes a Collision object (containing contact points, impulse, and relative velocity). OnTriggerEnter passes only the Collider reference of the object that entered.

Q3: Steps to load a scene from script, what are the pre-requisites for a scene to load?

Steps:

1. Include the namespace at the top of your script: using UnityEngine.SceneManagement;
2. Call the load method: SceneManager.LoadScene("SceneName"); (or use the scene index).

Pre-requisites:

- **Build Settings:** The scene **must** be added to the "**Scenes In Build**" list in the Build Settings window (File > Build Settings). If it is not in this list, Unity cannot find or load it at runtime.

Q4: What Time.timeScale is used for, what should we be cautious about when using it?

Use:

Time.timeScale is used to control the speed at which time passes in the game.

- 1.0 is normal speed.
- 0.0 pauses the game (specifically physics and frame-rate independent calculations).
- 0.5 creates slow motion.

Cautions:

1. **FixedUpdate vs Update:** Time.timeScale affects FixedUpdate and Time.deltaTime. However, it **does not** stop Update() from running. Code inside Update will still execute every frame even if the scale is 0.
2. **Unscaled Time:** If you want a pause menu to animate while the game is paused, you must use Time.unscaledDeltaTime or Time.unscaledTime.
3. **Persistence:** Time.timeScale is a global setting. If you set it to 0 to pause and then load a new scene, the new scene will also be paused unless you manually set it back to 1.0.